

Reg No. _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Fourth Semester B.Tech (Minor) Degree Examination June 2022 (2020 Admission)

Course Code: RAT282**Course Name: INTRODUCTION TO INDUSTRIAL AUTOMATION**

Max. Marks: 100

Duration: 3 Hours

PART A*(Answer all questions; each question carries 3 marks)*

Marks

- | | | |
|----|---|---|
| 1 | Define automation, enlist any 4 advantages of automation. | 3 |
| 2 | Describe about Detroit type automation. | 3 |
| 3 | Enlist any 6 selection criteria for a sensor in industrial applications. | 3 |
| 4 | The speed of a gear having 60 teeth is measured using a proximity sensor. The output of the proximity sensor is fed to a counter with a gating time of 1s. The counter indicates a value of 120. Calculate the speed at which the gear is rotating? | 3 |
| 5 | Discuss about the direction control of stepper motor. | 3 |
| 6 | Explain about the types of storage systems used in material handling. | 3 |
| 7 | Describe how a pressure relief valve limit the maximum operating pressure. | 3 |
| 8 | Explain cushioning in pneumatic cylinders. | 3 |
| 9 | Draw the ladder logic of NAND gate and EXOR gate. | 3 |
| 10 | List the advantages of PLC over electromechanical relay control. | 3 |

PART B*(Answer one full question from each module, each question carries 14 marks)***Module -1**

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|----|---|---|
| 11 | a) Explain about different types of automation. | 8 |
| | b) With a neat sketch explain about different types of automated transfer lines used in an industry. | 6 |
| 12 | a) Explain different types of FMS layout. | 8 |
| | b) Discuss about the ways of adding intelligence to an automated system. Also explain about the components of expert systems. | 6 |

Module -2

- 13 a) Describe the construction and working of LVDT with neat diagram. 7
 b) Explain about any 2 types of thermoelectric sensors. 7
- 14 a) Using neat diagrams, illustrate the working of optical encoders. 8
 b) Describe the process of force measurement using strain gauge. 6

Module -3

- 15 a) Draw and explain the block diagram of basic servo mechanism. Explain about the construction of DC Servo motors. 8
 b) Explain various design considerations in material handling. 6
- 16 a) With neat sketches explain adaptive control of machine tools. 8
 b) What do you mean by Automated Guided Vehicles (AGV)? Explain different types of AGVs. 6

Module -4

- 17 a) Design a hydraulic circuit for generating A+A-B+B- cylinder sequencing using Cascade Method. 7
 b) With a neat sketch show the working of electro-pneumatic converter. 7
- 18 a) Explain about the construction and working of direction control valve and flow control valve. 8
 b) Discuss about Relays, Solenoids and Timers. 6

Module -5

- There are three motors (M1, M2 & M3), having a separate start and stop push button. M1 must be running before starting of M2 and M3 motor. M2 and M3
- 19 a) can start and stop without affecting M1 operation. Only two motors can run at the time. Starting of the third motor will shut down all the outputs immediately irrespective of its input. Develop a ladder logic for the system. 8
 b) Briefly explain coordinate measuring machine. 6
- 20 a) Explain about various blocks in machine vision systems with neat diagrams. 8
 b) Describe various non-contact inspection methods. 6
